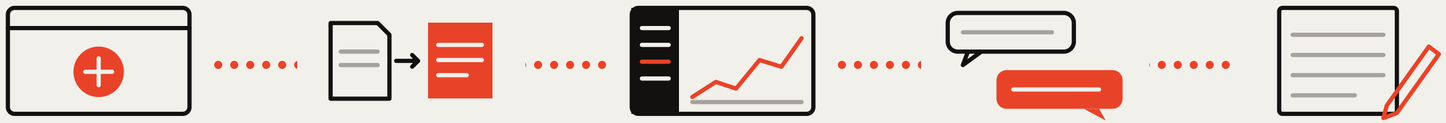
KaxaNuk
Sharing knowledge

— INVESTMENT BOOTCAMP

READY FOR PAPER TRADING? VALIDATE, FREEZE AND PRESENT.



06

From S05's finding to the decision: nothing wrong, the five criteria, frozen to watch it out of sample, and presented to a committee.

IN THIS SESSION

01 VALIDATE

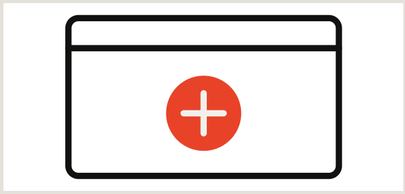
02 DECIDE

03 FREEZE

04 PRESENT

05 WRAP-UP

FROM YOUR FINDING TO YOUR DECISION.

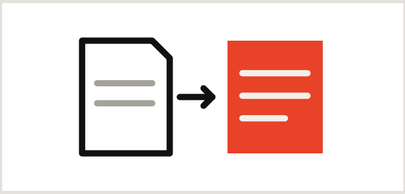


1 VALIDATE

That nothing is wrong.

- What cannot fail
- The blueprint judges the run
- The holdout, once

You leave with: your validation

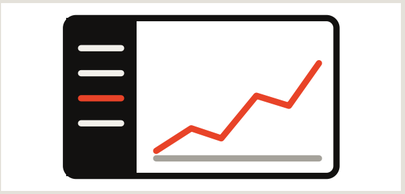


2 DECIDE

Five criteria. All of them.

- The gate, criterion by criterion
- Your book's verdict
- The signature is a person's

You leave with: your verdict

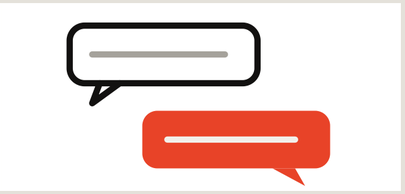


3 FREEZE

To watch it out of sample.

- The book, frozen as it is
- A test day first
- Every day, after the close

You leave with: your frozen book



4 PRESENT

To a committee.

- The rubric's nine rows
- Answer with a file
- Ten minutes

You leave with: your presentation



5 WRAP-UP

What you learned, and what comes next.

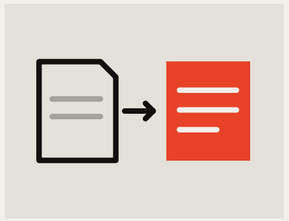
- The session's ideas
- Homework: close part H
- Next: the pillars

You leave with: your next step

EIGHT STEPS. IN THIS SESSION, THE SEVENTH.

S02 TO S05 | THIS SESSION | BEYOND THE COURSE

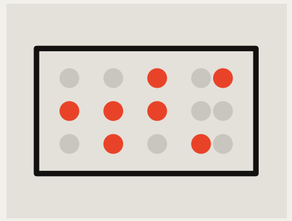
1 2 3 4 5 6 7 8



RESEARCH

S02

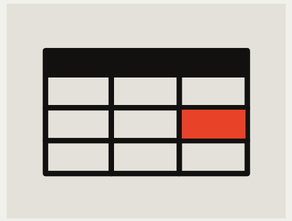
Your objective, from your library · OBJECTIVE.md



UNIVERSE

S02

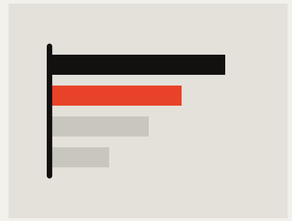
Which companies and when; delisted included · Universe/



DATA

S03

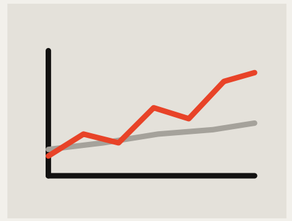
Your features, and whether they carry a signal · Data/



PORTFOLIO

S04

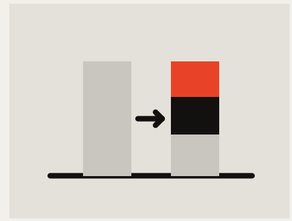
Your hypothesis, then the portfolios · Experiments/



BACKTEST

S05

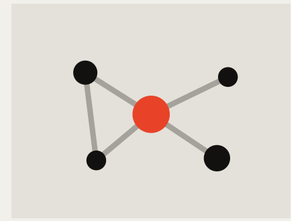
Your portfolio through time, net of costs



ATTRIBUTION

S05

Where the return came from



PAPER TRADING

S06

Tested without real money · Paper_Trading/

STEP 7



PRODUCTION

outside the course

From S05's homework

your `FINDINGS_1.md`, with your verdict, and N.

BEFORE DECIDING: WHAT CANNOT FAIL.

THE CHECK	WHERE IT SHOWS	FROM
The blueprint, saved first	Its version's time, before the rule.	S04
What was known that day	The signal at t – 1; the history before the date.	S04
The dead, included	The point-in-time seed; the coverage.	S02 · S03
Net, and every day priced	The complete run, with its two cost rows.	S05
N counted	Every signal and every book, in the journal.	S03 to S05
The holdout, once	Its prediction, saved before opening it.	S06
It reproduces	From a clean clone, with one command.	S06

IF ONE FAILS you do not decide: you fix it, and it counts.

THE BLUEPRINT JUDGES THE RUN.

ARYA · TREND-AND-VOLUME

`challenge`: my run against my blueprint.
1

Each prediction against its falsifier, with its file. Go ahead?

go ahead

Prediction 1 held. The benchmark did not.
2

THE PREDICTION, AGAINST THE RESULT · NET

NET SHARPE	VALUE	
The rule, with the 50/200 cross	0.30	
Its control, without the cross	0.25	
Margin over the control	+0.05	3
KN US Equity Core	0.41	

Prediction 1: more than +0.03 over its control. It held.

1. Ask for the challenge; read each line.
2. Each prediction against its falsifier.
3. The cross adds; the book does not beat the benchmark.

`challenge`

A WINNER MEANS LITTLE WITHOUT ITS N.

1999–2019

CORRECT FOR THE SEARCH

1999
SULLIVAN, TIMMERMANN & WHITE

THE UNIVERSE OF RULES
A trading rule, judged against all it was drawn from.

2000
WHITE

A REALITY CHECK
Does a search's best model beat its benchmark?

2015
HARVEY & LIU

THE HAIRCUT
A backtest's Sharpe, cut for the tests behind it.

WHAT THE SEARCH COSTS

2016
WIECKI, CAMPBELL, LENT & STAUTH

OUT OF SAMPLE
888 algorithms: the backtest Sharpe predicts little.

2017
SUHONEN, LENNKH & PEREZ

BACKTEST TO LIVE
215 strategies: Sharpe ratios fell once live.

2019
LÓPEZ DE PRADO

FALSE DISCOVERIES
Most are false: selection bias under many tests.

1

2014
BAILEY & LÓPEZ DE PRADO

THE DEFLATED SHARPE
Corrects for many tests and non-normal returns.

WHAT CHANGED

The best of many trials is the largest of many draws.

THEREFORE

Publish N. Then deflate the winner, with ①.

Each line from its paper's published abstract.

OPENED ONCE, AFTER ITS PREDICTION.

NET	THE RULE	ITS CONTROL	KN US EQUITY CORE
In sample, 2000 to 2024	0.24	0.20	0.38
Holdout, 2025 to 2026	1.04	0.98	0.93
Annual return, in sample	4.6%	3.4%	7.7%
Annual return, holdout	40.0%	35.0%	20.5%
Max drawdown, holdout	−28.3%	−27.3%	−19.4%

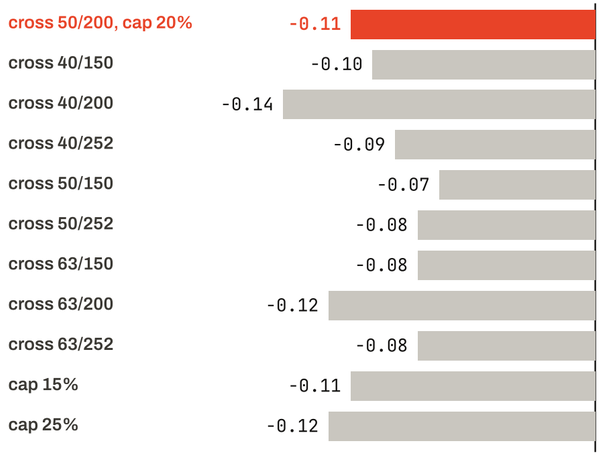
A YEAR AND A HALF little evidence: it confirms, it does not decide.

DOES IT HOLD IF YOU MOVE IT A LITTLE?

NET SHARPE, LESS THE KN US EQUITY CORE'S · N = 11

Sharadar · KN US Equity Core · provisional

EACH VARIANT, THE SAME RULE; THE CROSS OR THE CAP CHANGES



Net of 10 basis points per side, 2000 to 2026. No variant beats the benchmark: the curve is flat, and below zero.

A curve, not a peak

flat: the conclusion does not depend on 50/200.

FIVE CRITERIA. ALL OF THEM.

CRITERION	EVIDENCED BY
1 · Beats its benchmarks and its control, risk-adjusted	FINDINGS_1.md's table, on the same window.
2 · Idiosyncratic return in both layers	Brinson-Fachler and the factor model.
3 · Robust, with N published	A curve of variants; the trial count.
4 · Costs and capacity, modelled and stated	The cost rows; the capacity.
5 · A signature	A name and a date in BITACORA.md.

IF IT IS NOT EVIDENCED by FINDINGS_1.md and RESULTS.md, it is not met.

YOUR BOOK'S VERDICT.

CRITERION	THE EVIDENCE	
1 · Benchmarks	Sharpe 0.30 against the KN US Equity Core's 0.41; it beats its control.	No
2 · Idiosyncratic	Selection of +1%; Jensen's alpha, the intercept, negative.	No
3 · Robust	Eleven variants, a flat curve; N published.	Yes
4 · Costs	Costs modelled; capacity, not.	No
5 · Signature	Not sought: 1, 2 and 4 block.	—

IT DOES NOT PASS the blocking rows are the content: on to experiment 2.

REJECTED, WITH THE SAME CLARITY.

PAPER_TRADING/BITACORA.MD · CURRENT STATUS

```
## Current status
Experiment 1 · the 50/200 rule: not graduated.                    1
Blocks: 1 (below the KN US Equity Core), 2 (Jensen's alpha, the intercept,
  negative), 4 (capacity not modelled).
Clears: 3 (eleven variants, a flat curve; N published).
Criterion 5: not sought.
Next: Experiment 2, a new blueprint.                              2
```

1. A no, with the same shape as a yes.
2. A new idea is another experiment.

What failed, too

stays in the record: it is what you learned.

THE SIGNATURE IS A PERSON'S.

Criterion 5

A name and a date.

Your researcher

Lays out 1 to 4, evidenced or not, and stops.

It never declares

That the strategy graduates.

It never computes

A figure the files do not hold.

You sign

Or not, with the blocking rows.

The signature is sought when 1 to 4 are evidenced, never before.

FREEZE THE BOOK, AS IT IS.

PAPER_TRADING/PAPER_TRADING_1/

`paper_trading_1.py`	the rule, from the experiment's cells [1]
`FREEZE.json`	the commit, the date, each file with its hash, the versions [2]
`the refinery and its calculations`	copied, byte for byte
`the seed and the modules`	copied from the commit
`the security master`	from disk; git does not keep it

A changed file stops the book: what is frozen does not move.

Version 1.0.0: the first book on paper trading, reproduced from a clean clone.

1. Promoted, not copied: the same number.
2. promote.py, on a clean tree.

Never twice

a frozen book is never frozen again: another freeze is another book.

A TEST DAY BEFORE THE FREEZE.

In a scratch clone

Frozen with no commit; one day with `--dry-run`.

The membership

No members after the last file: it sells everything that day.

The seed

A span closed on its last day: it stops the live ones.

The new days

They are the first any code reads.

Both faults only show there: fix them before the real freeze.

EVERY DAY, AFTER THE CLOSE.

DAILY_UPDATE.PY · ONCE A DAY

```
the day's prices, once · checked                                  1
each book: its frozen refinery and rule
the engine three times: the history, after the window, since the freeze
the record: six tables; a restatement is flagged, never overwritten   2
it exits 0, 1 or 2: clean, flagged, failed                        3
a day with no price or an impossible move: it stops
the scheduled task, you register it
```

1. A changed file: `unfrozen-input`.
2. The past is not rewritten.
3. Read the flags, not only the exit code.

WHAT YOU WATCH OUT OF SAMPLE.

WATCH	WHY
Whether it behaves like its backtest	Turnover, holdings, exposure, costs: within their bands.
The kill switch	Written first: what would stop it.
The review dates	When it is read, and how long until production.
Nothing is refitted	A run that tunes anything is another backtest.

A FEW MONTHS DO NOT PROVE SKILL they prove the plumbing works.

TWO TRACKS, ONE FROZEN BOOK.

INVESTMENT LAB

NO CODE

Paper trading: your book →

its bands and its kill switch →

the record, every day

Frozen: the rule and its versions.

Every day: after the close.

Bands: from FINDINGS_1.md.

Kill switch: BITACORA.md's.

ARYA · TREND-AND-VOLUME

Signed: freeze book 1. 1

First a test day in a clone. Then promote.py and FREEZE.json. Go ahead?

go ahead

Frozen. The command to schedule it is yours: here it is. 2

1. Only after the signature.
2. You register the scheduled task.

```
paper-trading-gate
```

NINE ROWS. NONE FOR RETURN.

PRESENTATION DAY · A JUDGE'S SHEET

The 3x3 rubric · a row per file

AREA	ROW	A JUDGE OPENS	0 · 33 · 66 · 100
Strategy definition	Hypothesis and source of returns	OBJECTIVE.md	
	Research foundation	Bibliotheca/	
	Scientific process	git log	
Feature engineering	Data quality and integrity	Universe/	
	Feature design and relevance	RESULTS.md	
	Signal construction	c_* · r_*	
Strategy design	Portfolio construction logic	BLUEPRINT_1.md	
	Backtesting and robustness	FINDINGS_1.md	
	Attribution and understanding	FINDINGS_1.md	

reproduces | blueprint saved first | figures traced | N published | holdout once | gate evidenced

Six process gates beside the rows: pass or fail.

Equal weight: 0 not there · 33 basic · 66 clear · 100 rigorous, reproducible, documented. The handbook's reference pages: *Six questions, each to its row*; *Ten minutes, built from files*.

A LOSS CAN SCORE 100 when it is measured, counted and explained.

ANSWER A JUDGE WITH A FILE.

BREAKOUT ROOM · GOLDEN-FLOW FIRST

JUDGE · PORTFOLIO CONSTRUCTION LOGIC
What is your control?

`BLUEPRINT_1.md`, *Rules.*

JUDGE · BACKTESTING AND ROBUSTNESS
How many did you try?

`RESULTS.md`, *The trial count.*

YOUR PARTNER'S CARD

0 · 33 · 66 · 100	A JUDGE ASKS	FILE · SECTION
	What is your control?	
	How many did you try?	
	Which factor is it?	
	Net of what?	
	Who is missing?	
	What would kill it?	

Questions 1 to 3 on Golden-Flow, then swap for 4 to 6; then one each on your own repository. Share your screen; push nothing. Alone: your researcher asks, one question at a time; you answer with a file.

Each answer, a file and its section. Golden-Flow · RESULTS.md · FINDINGS_1.md · run 2026-10-06 · Backtest Engine 0.66.0 · Attribution Analysis 0.2.0 · hypothetical

TEN MINUTES BEFORE THE COMMITTEE.

MINUTE	WHAT YOU PRESENT	THE FILE
0 to 2	The hypothesis and why it should work.	OBJECTIVE.md
2 to 3	The universe and the benchmark, with the dead.	Universe/
3 to 5	The rule and its control, saved first.	BLUEPRINT_1.md
5 to 7	Net, against the control and the benchmark; N.	FINDINGS_1.md
7 to 8	Where it came from: factors and sectors.	FINDINGS_1.md
8 to 10	The verdict, and what comes next.	BITACORA.md

A NO IS PRESENTED TOO measured, counted and explained.

THE SESSION'S IDEAS, IN YOUR WORDS.

01

What do you check before deciding?

02

Why the five criteria, all of them?

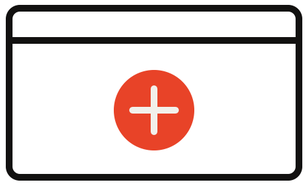
03

What is frozen, and why never twice?

04

What do you watch out of sample, and what not?

BEFORE THE COMMITTEE: CLOSE PART H.



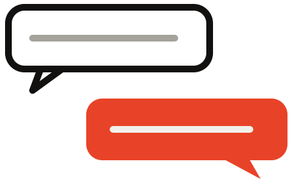
01 VALIDATE

- The list, check by check.
- Your holdout, once.
- Your curve of variants, with N.



02 DECIDE

- The five criteria, with their evidence.
- BITACORA.md: yes or no.
- If yes: the test day and the freeze.



03 PRESENT

- Ten minutes, with your files.
- Rehearse the six questions.
- Ask `next`.

VALIDATED FIRST. FROZEN NEXT.

Your strategy, validated, decided in writing and, if it passes, frozen to watch it out of sample.

LEARN

The pillars go deeper into each step.

`github.com/KaxaNuk/Investment-Bootcamp`

TEST

Your strategy on the Lab; universes and factors from Analytics.

`kaxanuk.mx/lab` · `kaxanuk.mx/analytics`

TELL US

What your committee decided.

QUESTIONS `workshops@kaxanuk.mx`

NEXT

The pillars: Coding Foundations, Financial Markets, Mathematical Finance

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Nothing here is a recommendation to buy, sell or hold any security; backtests are hypothetical, in sample and never traded; this session's figures are our own calculation on past data, not a result; paper trading uses no real money. Your researcher can be wrong; every decision is yours. What your researcher reads goes to the company behind your assistant, never to KaxaNuk.

Tools read 2026-10-13; works and links, 2026-10-11; the figures, from Sharadar prices of the KN US Equity Core, whose methodology is still provisional, computed 2026-10-17, with 10 basis points per side. Shared under CC BY 4.0; the KaxaNuk name, logo and isotipo are not.